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DSC680 – Applied Data Science
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10 Questions I anticipate:

1. How can one distinguish between genuine and fabricated news?
    a. The best indicator is the language used in the title. Inflammatory all caps language is commonly associated with fake news. Domains/URLs are also indicative of fake news resources.
2. Would this intervention hinder the freedom of expression of users?
    a. Not at all! This dataset was explicitly trained on the posting of articles. If a user wanted to post a fake news article, their account and post would be flagged, but if they want to share their opinions, that would still be allowed.
3. What are the potential risks associated with the dissemination of fabricated news?
    a. Fake news can lead to all sorts of societal ills, such as violence, erosion of trust, pitting groups against each other.
4. Should the government take action to address this issue?
    a. Our models can be trained without government intervention. The possibility of the government weaponizing this technology and targeting opposing viewpoints as “fake news” is worrisome. If social media websites such as Twitter and Facebook are unwilling to implement these changes, it could be regulated.
5. What level of accuracy can be anticipated in distinguishing between authentic and fabricated news?
    a. We can predict with up to 90% accuracy whether a news article is fake or real.
6. Why has this problem not been resolved yet?
    a. There is a financial incentive to allow salacious fake news on social media. The content is highly engaging and can be monetized. Social media websites need to be pushed by their users or by governments to crack down on social media if they won’t do some themselves.
7. Is the situation improving or deteriorating?
    a. Despite advancements in technology, fake news is still quite common. The main problem today is that fake news is not a priority when content is posted. All in all, the situation is deteriorating.
8. Is fabricated news specifically targeting a particular demographic group?
    a. Fake news can be biased against any groups. It plays into fears and desires of the readers. Based on a large number of fake news originating from countries such as Russia, China, and Iran, foreign influence is

attempting to erode the unity within the United States and the West as part of a regional power play.

9. Which models were selected and for what reasons?
    a. I chose 3 different models, SVC, logistic and gradient boost. Based on the recommendations from Elsa Norman, these models were best suited for the classification test.
10. Which dataset was chosen and for what reasons?
    a. I chose an open-source dataset from Kaggle. First, it was best suited for the problem as it contained the data along with “fake/real” labels. Second, the dataset I chose had already been downloaded 10,000 times. Think of this as the data being “peer reviewed” 10,000 times. Finally, the dataset was compiled from PolitiFact fact checkers, so each item had already been proven true or false and could be traced back to the PolitiFact website.